

Chapter 7

LUBRICANTS

7.1 Introduction

A lubricant is a substance introduced between two moving surfaces to reduce friction. This property of reducing friction is termed *lubricity* or *slipperiness* and the process of reducing friction between two moving surfaces by the introduction of lubricants is known as *lubrication*. Before going into what lubrication is, what a lubricant is and how it functions, we need to know what friction is. Friction is the force that resists the relative motion between two bodies in contact. The surfaces of all material, no matter how smooth they appear, are not truly smooth on an atomic scale. They are rough, with sharp and rugged projections called asperities (Fig. 7.1(a)). When two surfaces come in contact with each other, initially, they only touch at a few of these asperity points (Fig. 7.1(b)). When a small load is applied on a metal surface, local high pressure develops at these points (asperities) of contact, which then tend to interlock, thus developing a resistance to motion. When a metal slides, the peaks get broken resulting in wear and tear of the surfaces. Further, if the pressure is high, the asperities deform plastically increasing the contact area between the two surfaces. A sufficiently high pressure leads to the formation of a welded junction between asperities. Additional force is then required to break these junctions. This further adds to the frictional resistance.

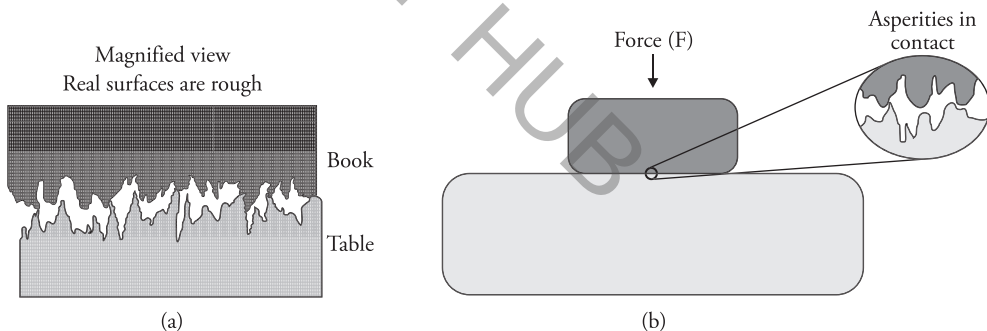


Figure 7.1 (a) Magnified view of a real surface. (b) Contact between two surfaces on application of load

Thus, frictional resistance harms the system in three ways.

- It causes energy loss. Extra work has to be done to overcome the force of friction.
- Friction causes generation of heat in areas in contact. These heated parts may melt and get deformed, leading to seizure (welding of two surfaces due to heat) of the system.
- Breaking of asperities brings about abrasion or surface wear and tear.

7.2 Functions of Lubricants

1. A lubricant reduces frictional resistance by keeping the moving surfaces of a system apart and prevents their direct contact so that the asperities do not interlock.
2. It reduces surface deformation, and wear and tear by keeping the moving surfaces apart.
3. The lubricating oil absorbs the generated heat and acts as a coolant.
4. Lubricants increase the efficiency of the machine.
5. They reduce maintenance cost of machines and increase their life.
6. Lubricants also act as sealants for gases.
7. Lubricant circulation systems carry away debris and contaminants that get introduced into the system.
8. They reduce noise of the machine.
9. They avoid seizure of the moving parts since the use of lubricants minimize the liberation of frictional heat.
10. Lubricants prevent corrosion and rust.

7.3 Mechanism of Lubrication

Depending upon the operating conditions and the lubricant characteristics, lubrication is of the following types:

1. Fluid film or thick film or hydrodynamic lubrication
2. Thin film or boundary lubrication
3. Extreme pressure lubrication

Fluid film or thick film or hydrodynamic lubrication

Hydrodynamic lubrication occurs when the load is low and the speed is high. In this type of lubrication, the two sliding metal surfaces are separated by a thick, continuous unbroken film or layer of the lubricant. The thickness of the fluid film is at least $1000 \text{ \AA}$. The two moving surfaces do not come in contact with each other and the load is taken completely by the oil film. Since the metal surfaces are not in contact with each other, the friction reduces. Whatever friction is there is due to the viscosity or internal resistance of the lubricant. The coefficient of friction in such cases is

as low as 0.001 to 0.03 as compared to 0.5 to 1.5 for sliding metal to metal surfaces in the absence of lubrication.

The basis of hydrodynamic lubrication is the formation of an oil wedge or hydrodynamic wedge between the two moving surfaces. For example, consider the rotation of a shaft with respect to a stationary bearing. The lubricant is introduced in the annular space. In the rest position, the shaft rests at the bottom of the bearing and all the lubricant is squeezed out, resulting in contact between the two surfaces (Fig. 7.2). As torque is applied to the shaft, the fluid pressure forces the surface apart (due to hydrodynamic lift) and the hydrodynamic or oil wedge is developed. Consequently, one surface floats over the other surface without mutual contact.

The lubricants chosen for hydrodynamic lubrication should have minimum viscosity and at the same time remain in its place and separate the sliding surfaces. Hydrocarbon oils are considered to be satisfactory lubricants for fluid film lubrication. However, their viscosity decreases with rise in temperature. Therefore, to maintain their viscosity in all seasons of the year, they are blended with suitable long chain polymers. Moreover, hydrocarbon oils contain a small fraction of unsaturated hydrocarbons which tend to get oxidized forming gummy products. Hence, antioxidants such as aminophenol are also added.

Examples of systems which use hydrodynamic lubrication include the following.

- Delicate instruments like watches and clocks.
- Light machines like sewing machines and scientific instruments.
- Large plain bearings like pedestal bearings and main bearings of diesel engines.

Elastohydrodynamic lubrication Elastohydrodynamic lubrication is similar to hydrodynamic lubrication, but it occurs when the surfaces are in rolling motion relative to each other, like in ball bearings and roller bearings. In elastohydrodynamic lubrication, the pressure is higher and the thickness of the film is much less than hydrodynamic lubrication. It is called elastohydrodynamic lubrication because the film elastically deforms the rolling surface to lubricate it.

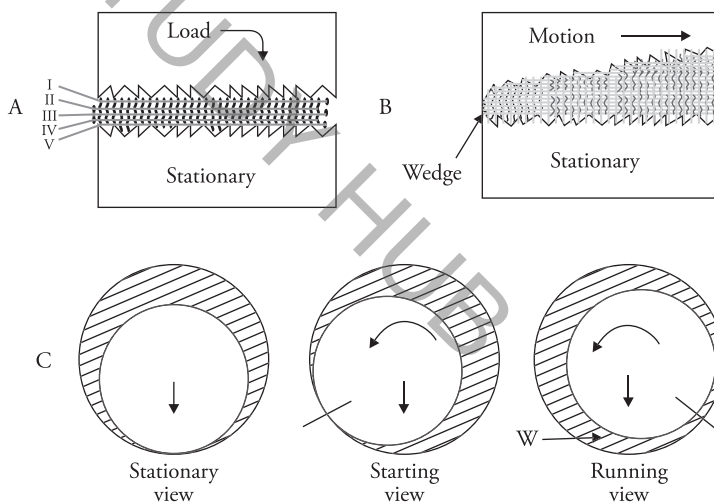


Figure 7.2 • Hydrodynamic lubrication

Thin film or boundary lubrication

Boundary lubrication occurs in systems under high load and slow speed. In this kind of lubrication, the lubricating film is thin. It has about the same thickness as the surface roughness such that the high points or asperities on the solid surface are in contact. Figure. 7.3 (a) the load is carried by the surface asperities rather than by the lubricants. The oil film thickness is so small that the effectiveness of the lubricant depends on its oiliness, which is the ability of the lubricant to stick on the surface. The thin film of the lubricant must be adsorbed on the metal surface by physical and chemical forces. Sometimes, the lubricants react chemically with the metal surface forming a thin film of metal soap which acts as a lubricant.

In this type of lubrication, the moving surfaces are not completely separated. However, the points of contact are considerably decreased resulting in lowering of frictional resistance. The thickness of the boundary film may be one or two molecular layers of thickness 20–30 Å (Fig. 7.4). The coefficient of friction in this case is generally 0.05 to 0.15. This friction depends upon the structure and chemical properties of the oil used for lubrication.

Boundary lubrication is undesirable since the metal surfaces are in contact which leads to increased friction, energy loss, wear and tear, and material damage. However, it is inevitable during the operating life of a machine especially when

- A shaft starts moving from rest
- Speed is very slow
- Load is high
- During shut down
- Viscosity of the lubricant is very less
- Under conditions of shock, jerk, etc.

For boundary lubrication, the lubricant should have the following characteristics.

- Its oiliness or the capacity to stick to the metal surface should be good.
- It should have long hydrocarbon chains which increases the lateral attraction between the chains (larger the hydrocarbon chain, greater are the van der Waals forces of attraction).
- It should have polar groups and active groups which adsorb on the metal surface or react chemically with it.
- It should have high-viscosity index and resistance to oxidation.
- Cloud and pour point should be low.

Vegetable oils, animal oils and glycerides of higher fatty acids, and their soaps have good oiliness, for example, oleic acid ($C_{17}H_{33}COOH$) and stearic acid ($C_{17}H_{35}COOH$). Their polar $-COOH$ group reacts with the metal surface to form a continuous film and the hydrocarbon chains of fatty acids are oriented outwards in a perpendicular direction (Figure 7.4)

Graphite and MoS_2 are also used for boundary lubrication. They can bear compression as well as high temperatures. These solid lubricants can be used either alone or as a stable suspension in oil.

Examples of systems using boundary lubrication include the following.

- Lubrication of journal bearing in diesel engines (especially during the starting and stopping of engines).
- Guide and guide shoe in two stroke engines.
- Hydraulic pumps such as the vane pump.
- Valve trains, especially cams and followers.

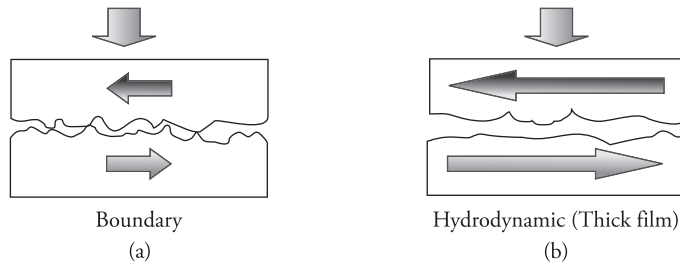


Figure 7.3 (a) Boundary lubrication and (b) Hydrodynamic lubrication compared

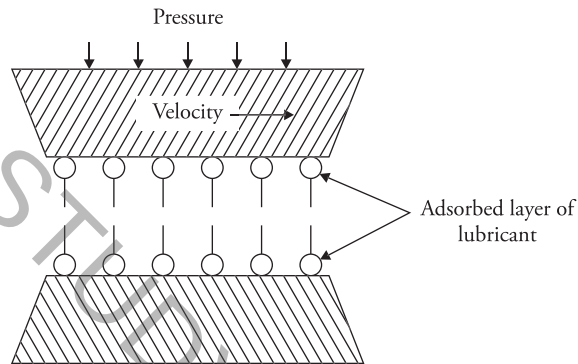


Figure 7.4 Hydrocarbon chains of fatty acids oriented outwards in a perpendicular direction in boundary lubrication

Extreme pressure lubrication

When the moving surfaces are subjected to very high pressure and speed, heat is generated causing the following harmful effects to occur.

- Local high temperature develops resulting in the formation of a welded junction and metal tearing.
- The lubricants fail to stick to the metal surface. They vapourise or decompose.
- Surface deformation and seizure takes place.

To overcome the above difficulties due to extreme pressure conditions, extreme pressure additives are added to the lubricant. The commonly used extreme pressure additives are organic compounds having active radicals or groups such as chlorine (as in chlorinated esters), sulphur (as in sulphurized oils) or phosphorus (as in tricresyl phosphate). The compounds interact with the metal surface at the molecular level to form a protective compound which increases the thermal stability of the metal surfaces. The compounds are generally metallic chlorides, sulphides or phosphides which have high melting points. For example, when additives are added to a lubricant applied on iron and steel, they form iron chloride (MP: 649 °C) or iron sulphide (M.P: 1170 °C). It is important to note that under extreme pressure conditions, the lubricant is not just a film or coating over the metal surface, it is a permanent modification of the metal surface. The surface compound is a few molecule layers thick. The coefficient of friction is 0.1 to 0.4 cm/s. These films remain intact under high pressure and temperature and lubricate the surface. If by chance the film breaks due to the rubbing action of the moving parts, it is immediately replenished. Extreme pressure lubricants find use in aircrafts and spacecrafts.

7.4 Classification of Lubricants

On the basis of their physical state, lubricants can be broadly classified into the following.

1. Liquid lubricants or lubricating oils
2. Semi-solid lubricants or greases
3. Solid lubricants

Liquid lubricants or lubricating oils

Liquid lubricants reduce friction and wear and tear between two metal surfaces by providing a continuous layer between them. They also act as a cooling medium, prevent corrosion and act as a sealing agent. Liquid lubricants can be further subdivided into three types.

- (i) Animal and vegetable oils
 - (ii) Mineral oils or petroleum oils
 - (iii) Blended oil, doped oils or compound oils
- (i) **Animal and vegetable oils** These are extracted from animal and vegetable matter and contain glycerides of higher fatty acids. They are known for their 'oiliness', a property by which they stick to the metal surface even under high temperature and high load. However, they have certain limitations. They are expensive; undergo oxidation to form gummy and acidic products; they oxidize in contact with air and become thick; and they hydrolyse in the presence of moisture in air. They are rarely used for lubrication these days because of these limitations. Rather, they are used as blending agents in mineral oils to improve the oiliness of the latter.

The commonly used animal and vegetable oils are olive oil, mustard oil, castor oil, palm oil, rapeseed oil, whale oil etc.

- (ii) **Mineral oils or petroleum oils** These oils are obtained by the fractional distillation of crude petroleum. On being subjected to vacuum distillation, the heavy residual fraction yields lubricating oil. The hydrocarbon chain is 12 to 50 carbon atoms long. They are widely used as lubricants because they are cheap, abundant and stable under working conditions. They are not readily hydrolysed. However, they have poor oiliness as compared to vegetable oils. Their oiliness is increased by blending them with vegetable oils.
- (iii) **Blended oils** Blended oils are termed as doped oils or compounded oils. No single oil can serve as a satisfactory lubricant under all conditions. Hence, their properties are improved by adding certain additives. This phenomenon is called 'blending'. The different additives employed are as follows.
- (a) **Antioxidants or inhibitors** They retard the oxidation of lubricating oils by getting themselves preferentially oxidized. Antioxidants are added in lubricants used in internal combustion engines. Examples include aromatic phenolic or amino compounds.
 - (b) **Oiliness carriers** They increase the oiliness of the lubricant. These carriers are added in lubricants used in boundary lubrication. Examples include vegetable oils like castor oil, mustard oil and fatty acids like oleic acid, palmitic acid, stearic acid, etc.
 - (c) **Viscosity index improvers** They increase the viscosity index and are added in lubricants used in machines working under a broad temperature range. Example includes hexanol.
 - (d) **Pour point depressants** They reduce the pour point of the oil. Examples include alkylated naphthalene, phenolic polymers and polymethacrylates.
 - (e) **Detergents and deflocculents** They neutralise sludge and varnish precursors and keep the surface deposit free by maintaining them in soluble form. Examples include calcium and barium salts of phosphates and sulphonates.
 - (f) **Dispersants** They prevent and retard sludge formation. Dispersants are added in lubricating oils used in internal combustion engines. Examples include high-molecular-weight amides and amide polymers.
 - (g) **Metal deactivators** They stop the catalytic effect of metals on the oxidation states. Examples include amines, sulphides or phosphides.
 - (h) **Antifoaming agents** They prevent the formation of stable foams. Examples include glycol, glycerol and silicon polymers.
 - (i) **Rust and corrosion inhibitors** They protect the metal from corrosion, for example, organometallic compounds, metal phenolates, basic metal sulphonates, fatty acids and amines.
 - (j) **Extreme pressure additives** They enable the lubricant to function under the conditions of high temperature and pressure, for example, sulphurized fats, phosphorus compounds, etc.
 - (k) **Abrasion inhibitors** They provide abrasion resistance, for example, tricresyl phosphate.
 - (l) **Emulsifiers** They help in the formation of emulsions of lubricating oils with water, for example, sodium salts of sulphonic acids and sodium salts of organic acids.

Semi-solid lubricants

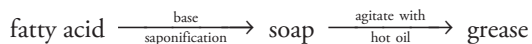
Greases are the most common examples of semi-solid lubricants. They are oils made semi-solid by adding a thickening agent. The thickeners dispersed in the lubricating oil produce a stable and colloidal structure or gel. In fact, greases contain three basic active ingredients – a lubricating oil, additive and thickeners (Table 7.1). Thickeners are mainly special soaps of lithium, sodium, calcium, barium, etc. Non-soap thickeners like carbon black, clay, asphaltenes and siloxane are also used.

Table 7.1 Basic ingredients of grease

Lubricating oil	Thickener	Additive
Petroleum oil Polyglycols Polyalpha olefins Vegetable oils	Soap of Li, Na, Ba, Ca, Al Non-soap (inorganics) clay, graphite, silica, carbon black	Antiwear, antioxidant, rust and corrosion resistant, viscosity index improver, pour point depressers, extreme pressure additives, etc.

Preparation of greases

Greases are prepared by the *saponification* of fat (such as tallow or fatty acid) with alkali (NaOH, Ca (OH)₂, etc.) followed by the addition of hot lubricating oil with constant stirring



The consistency of the grease depends upon the quantity of oil added. Mineral oil plays an important role in determining the performance of the grease. The nature of soap used decides the consistency of the grease, the temperature up to which it can be used and its resistance to water and oxidation. Greases are classified on the basis of the soap used in their manufacture (Table 7.2).

Table 7.2 Classification of greases

Type of grease	Chemical composition	Characteristic properties
Calcium base greases	Emulsions of petroleum oils with calcium soaps	Insoluble in water hence water resistant. Since oil and soap separate above 80 °C, these greases are used up to 80 °C. They are suitable for water pumps, tractors, etc.
Soda-based greases	Petroleum oils, thickened by mixing sodium soap	Not water resistant because sodium content of soap is soluble in water; they can be used up to 175 °C; they are suitable for use in ball bearings where the lubricants gets heated due to friction
Lithium-based greases	Petroleum oil thickened by mixing lithium soap	Water resistant but can be used up to 15 °C only. They are expensive
Axle greases	Resin and fatty oils mixed with lime or any heavy metal hydroxide (talc and mica are added as fillers)	Water resistant, cheapest of all greases; they find use in equipments working under high loads and low speeds
Non-soap greases	Mineral oil with solids like graphite, carbon black and soapstone	Used in rail axle boxes, machine bearings, tractors, rollers, etc.

Applications

1. Greases are generally used in textile mills, paper mills and mills of edible articles where use of liquid lubricants can cause contamination of the products by spilling or dripping.
2. Machines like rail axle boxes work at slow speed and high pressure, intermittent operation and sudden jerks. In such conditions, oil cannot remain in its place.
3. In bearing and gears working at high temperatures.
4. Used in bearings which need to be sealed against the entry of dust, grit and moisture.

However, there are certain limitations of greases. These are as follows.

1. Their coefficient of friction is much higher than that of lubricating oils; therefore, wherever possible lubricating oils are preferred over greases.
2. Compared to lubricating oils, greases cannot dissipate heat effectively from the bearing.
3. Greases cannot be used in the presence of water because the fibre structure of the thickeners is adversely affected by water and the grease undergoes degradation.

Solid lubricants

Solid lubricants are used in places where liquid and semi-liquid lubricants fail to give satisfactory results. They offer lubrication at

- Very high temperatures and pressure where other lubricants fail to stay in their position.
- Solid lubricants like MoS_2 are used in vacuum where liquid and semi-solid lubricants are ineffective.
- Solid lubricants are used in industries where entry of dust is not acceptable.

The most common solid lubricants are graphite and molybdenum disulphide. Their lubrication properties are due to their layered or lamellar structures. The layers are held together by weak van der Waals forces and hence can easily slip over each other.

- (i) **Graphite** Graphite consists of a stack of carbon atom sheets, each sheet containing a hexagonal arrangement of atoms. Each carbon is sp^2 hybridised. The adjacent layers are separated by a distance of 3.4 Å and are held together by weak van der Waals forces. Hence, graphite is soapy and slippery by nature. It is non-inflammable and is not oxidized in air below 375 °C. Graphite is used either as a powder or as a suspension or as grease. In dry form, it is an effective lubricant additive because of its lamellar structure. The lamellas orient parallel to the surface in the direction of motion. It can also be used as a suspension with oil or water. Its suspension in oil is called 'oildag' (*dag* meaning deflocculated adhesion graphite). Oildag finds use in internal combustion engines because it provides a tight fit contact by forming a film between the piston and the rings. Suspension of graphite with water is called 'aquadag'. It is prepared using tannin as an emulsifying agent. Aquadag is used where oil free lubricant is required.

Graphite is best suited for lubrication in air. Water vapour is essential for graphite lubrication. Water is adsorbed; it reduces bonding energy between the hexagonal planes of graphite. Graphite cannot be used in vacuum because it needs water vapour to lubricate (for figure, refer to Fig. 11.48, Chapter Solid State).

Applications Graphite is commonly used in air compressors, food industry, ball bearings, railway track joints, etc.

- (ii) **Molybdenum disulphide (MoS_2)** It has a sandwich like structure in which the molybdenum atoms lie between two layers of sulphur atoms (Fig. 7.5). MoS_2 has low shear strength in a direction parallel to the layers due to poor interlaminar attraction. Hence, it has low coefficient of friction. It is stable in air up to 400°C . MoS_2 has excellent adhesion, high load carrying capacity and protection against corrosion. However, it cannot be used in wet environment as moisture increases friction. MoS_2 is an excellent lubricant in vacuum, hence, it finds use in space vehicles. A solid film lubricating surface useful for space vehicles is made from 70% MoS_2 + 7% graphite bonded with 23% silicates. It can withstand extreme temperatures, low pressure and nuclear radiations.

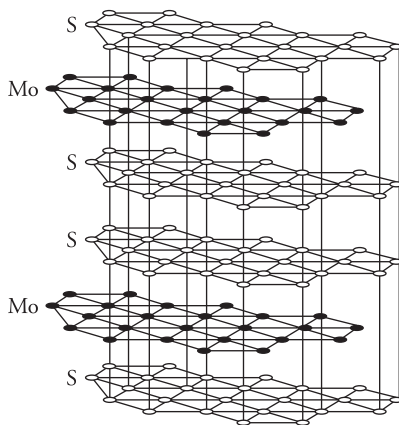


Figure 7.5 Sandwich like structure of MoS_2

Besides graphite and MoS_2 , other common solid lubricants are soapstone, talc, mica, polytetrafluoroethylene (PTFE), hexagonal boron nitride (called white graphite and is used in space vehicles), etc.

7.5 Synthetic Lubricants

Conventional lubricants obtained from natural oils are a cocktail of hydrocarbons and in spite of aggressive solvent base refining, they contain many organic as well as inorganic impurities. This causes oxidation, acid formation and promotes the formation of sludge in the lubricants.

Synthetic lubricants are engineered products. They are synthesized by chemical reactions. All the components are of high purity and as a result, the end product formed is a pure compound which is less vulnerable to oxidation and breakdown. It does not emulsify and produce undesirable products. Synthetic lubricants are tailor made and have a unique combination of properties like the ability to perform satisfactorily over a wide range of temperature, high-viscosity index, resistance to hydrolysis, long service life even in reactive environment, etc.

The commonly used synthetic lubricants are as follows.

- (i) **Dibasic esters** di-2-ethyl hexylsebacate is the most widely used dibasic ester. It has a satisfactory performance from $(-50\text{ }^{\circ}\text{C}$ to $230\text{ }^{\circ}\text{C}$. They have low volatility, a high-viscosity index and high temperature stability and are non-corrosive, non-toxic and stable to hydrolysis. These are generally used in jet engines and military equipments.
- (ii) **Halogenated hydrocarbons** Although halogenated hydrocarbons have a low-viscosity index, they are good for extreme pressure lubricating properties. However, they are costly. They find use in submarines due to their low inflammability, resistance to oxygen, H_2O_2 and chemicals.
- (iii) **Polyglycols and their derivatives** They can be water soluble or insoluble depending upon the length of the carbon chain. Common examples are polyethylene glycol, polypropylene glycol, higher polyalkylene oxides, polyglycidyl ethers, etc.

Polyglycidyl ether is water insoluble but can adsorb water. These lubricants have low pour points, high-viscosity index, do not affect rubber and other gasket materials and are cheap. Polyethylene oxides however undergo thermal decomposition at high temperature and hence cannot be used at elevated temperatures. These lubricants find use in combustion engines, gears, compressors and pumps.

- (iv) **Organophosphate esters** They are fire-resistant and are used for extreme pressure lubrication.
- (v) **Silicones** They are excellent synthetic lubricants because of their resistance to oxidation up to $200\text{ }^{\circ}\text{C}$, high-viscosity index, chemical inertness, etc. Silicon fluids are generally used for rubber and plastic surfaces such as bushings, bearings, etc. They are also used as moisture repellent lubricants for clocks, timers and other electronic devices.
- (vi) **Organic amines, imines and amides** They possess low pour point, high-viscosity index and can be used in the temperature range $(-50\text{ }^{\circ}\text{C}$ to $250\text{ }^{\circ}\text{C}$.

7.6 Lubricating Emulsions

An emulsion is a colloidal solution consisting of more than one immiscible liquid dispersed in another in the form of droplets (for details, refer to the chapter on Surface Chemistry Section 14.14).

Emulsion (oil in water or water in oil type) can also be used for lubrication. In machining operation like milling, boring, turning, cutting, the tool employed gets heated to a high temperature. This may cause oxidation and subsequent corrosion of metal. In order to prevent overheating, effective cooling and lubrication is required. Oil in water emulsions are used as cooling and lubricating liquids in cutting tools. The oil lubricates and the water acts as a coolant. Fatty acids such as lard oil and sperm oil are very good for cutting tools. Water in oil emulsions are prepared by mixing water and oil containing 1 to 10% of water insoluble emulsifiers like alkaline earth metal soaps. They find use as compressors and pneumatic tool lubricants as the water evaporates and gives an internal cooling effect.

7.7 Properties of Lubricants

Viscosity

Viscosity is the property of a liquid or fluid by virtue of which it offers resistance to its own flow. A liquid in a steady state of flow on a surface may be supposed to consist of a series of parallel layers moving one above the other. Each layer moves faster than the lower layer due to internal friction.

Consider two layers of a liquid which are at a distance d , moving with a velocity difference v . Then, the force per unit area (F) required to maintain this velocity difference is given by

$$F = \frac{\eta v}{d}$$

where η (eta) is the proportionality constant known as the coefficient of viscosity, This constant η is characteristic of a particular liquid at a given temperature and decreases with increase in temperature. If $v = 1\text{ cm/s}$ and $d = 1\text{ cm}$, then

$$F = \eta,$$

Hence, coefficient of viscosity (η) may be defined *as the force per unit area required to maintain a velocity difference of unity between two parallel layers separated by unit distance.*

Absolute viscosity It may be defined as the tangential force per unit area required to maintain a unit velocity gradient between two parallel layers. Absolute viscosity is denoted by η (eta). Its unit in CGS system is poise. Poise being a bigger unit, centipoise is generally used. One centipoise = 0.01 poise.

Absolute kinematic viscosity It is the ratio of absolute viscosity to density of any fluid. Absolute kinematic viscosity is denoted by V . Its unit in CGS system is stoke. The smaller unit centistoke is commonly used. 1 centistoke = 0.01 stoke.

Centistoke = centipoise / density

Significance of viscosity Viscosity is a very significant property used to determine the quality of a lubricant under a given set of operating conditions. If the machine is moving at a slow speed under heavy load, lubricants of high-viscosity are preferred because low-viscosity lubricants will be squeezed out of the moving parts. On the other hand, in light machines like sewing machines, clocks, etc, low-viscosity lubricants are selected as high-viscosity oil will cause jamming of the machine parts.

Effect of temperature on viscosity Viscosity of a lubricant decreases with the rise in temperature. Hence at high temperature, the lubricant might be squeezed out of moving parts. Therefore, it is essential that the viscosity of the chosen lubricant should not vary much with the change in temperature – in this way, the lubricant remains in its place (between the moving parts) and can be used continuously at high temperatures as well. The rate of change of viscosity with temperature is measured on an arbitrary scale known as the *viscosity index*.

Viscosity index (VI) It is an empirical scale used to measure the change of viscosity of oil with temperature. The VI is defined as the average decrease in viscosity of oil per degree rise in

temperature between 100 °F and 210 °F. If the viscosity of a lubricant decreases rapidly with the rise in temperature, it is said to have low-viscosity index. On the other hand, if the viscosity does not change much with the increase in temperature, then the lubricant is said to have high-viscosity index.

Determination of viscosity index

Two types of standard oils are used to determine VI.

- Paraffinic-based Pennsylvanian oils whose viscosity does not vary much with temperature. These are known as H-oils and have been assigned VI = 100.
- Naphthenic-based Gulf oils which show a large change in viscosity with temperature. These are known as L-oils and have been assigned VI = 0.

To determine the viscosity index of a test oil

- The viscosity of the test oil is found experimentally at 100 °F and 210 °F. Let it be U and V , respectively.
- From the list of H-oils, that oil is selected whose viscosity at 210 °F is the same as that of the test oil at 210 °F and its corresponding viscosity at 100 °F is read off. Let it be H .
- Similarly, from the list of L-oils, that oil is selected whose viscosity at 210 °F is the same as that of the test oil at 210 °F and again its corresponding viscosity at 100 °F is read off. Let it be L .

VI is given by the formula

$$VI = \frac{L - U}{L - H} \times 100$$

where U = Viscosity of the test oil at 100 °F (38 °C)

H = Viscosity of H-oil at 100 °F having the same viscosity at 210 °F (99 °C) as the viscosity of the test oil at 210 °F.

L = Viscosity of L-oil at 100 °F. Viscosity of test oil and L-oil at 210 °F is the same.

The viscosity index depends directly on the molecular structure of the compound. Linear molecules have high-viscosity index. Branched molecules have low-viscosity index because branches prevent the alignment of the neighbouring molecules.

The viscosity index of lubricating oil can be increased by adding organic polymers like polyisobutylene, n-hexanol, etc. These polymers are partially soluble in the lubricant at low temperatures. The solubility of the polymer increases with the rise in temperature. Therefore, as temperature increases, two opposite effects occur.

- (i) Viscosity of the lubricating oil decreases.
- (ii) Solubility of the polymer in the lubricating oil increases, resulting in increase in its viscosity.

Because of these two opposing effects, the viscosity of the lubricant remains constant or in other words, the viscosity index of the lubricant increases.

Solved Examples

1. An oil sample under test has the same viscosity as that of the standard Gulf oil (low-viscosity standard) and Pennsylvanian oil (high-viscosity index standard) at 210 °F. Their viscosities at 100 °F are 61, 758 and 420 s, respectively. Calculate the viscosity index of the oil sample.

Solution

$$L = 758 \text{ s}, H = 420 \text{ s}, U = 61 \text{ s}$$

$$VI = \frac{L - U}{L - H} \times 100 = \frac{758 - 61}{758 - 420} \times 100 = 206.2$$

Practice problems

1. An oil of unknown viscosity index has a redwood viscosity of 64 s at 210 °F and 564 s at 100 °F. The low-viscosity standard (Gulf oil) possess a redwood viscosity of 64 s at 210 °F and 774 s at 100 °F. The high-viscosity standard (Pennsylvanian oil) has a redwood viscosity of 64 s at 210 °F and 414 s at 100 °F. Calculate the viscosity index of the unknown oil. [Ans 58.33]
2. An oil of unknown viscosity index has a viscosity of 60 s at 210 °F and 600 s at 100 °F. The high-viscosity index standard (i.e., Pennsylvanian oil) has a viscosity of 60 s at 210 °F and 500 s at 100 °F. The low-viscosity index standard (Gulf oil) has a viscosity of 60 s at 210 °F and 800 ss at 100 °F. Calculate the viscosity index of the unknown oil. [Ans 66.67]

Measurement of viscosity

The instrument used to measure viscosity is called a viscometer. Different types of viscometers are in use. They all work on the principle of allowing a fixed volume of oil to flow from a standard orifice at a fixed height and measuring the time of flow in seconds. The instruments commonly used are as follows.

- **Redwood viscometer** A Redwood viscometer measures the time required for 50 mL of the oil to pass through its orifice.
- **Saybolt viscometer** A Saybolt viscometer measures the time required for 60 mL of the liquid to pass through its orifice.

The viscosity determined by the above two methods is expressed in redwood seconds and saybolt universal seconds, respectively.

A brief description of a Redwood viscometer which is commonly used in our country is given here.

Based on the size of the orifice, Redwood viscometers are of two types – Redwood viscometer no. 1 and Redwood viscometer no. 2. Redwood viscometer no 1 with an agate jet of diameter 1.62 mm and length 10 mm is used for measuring the viscosity of thin lubricating oils. On the other hand, Redwood viscometer no. 2 is used for measuring the viscosity of thick lubricating oils like fuel oils, and has a jet of diameter 3.8 mm and length 15 mm.

Construction of a Redwood viscometer no. 1

It consists of the following parts (Fig. 7.6).

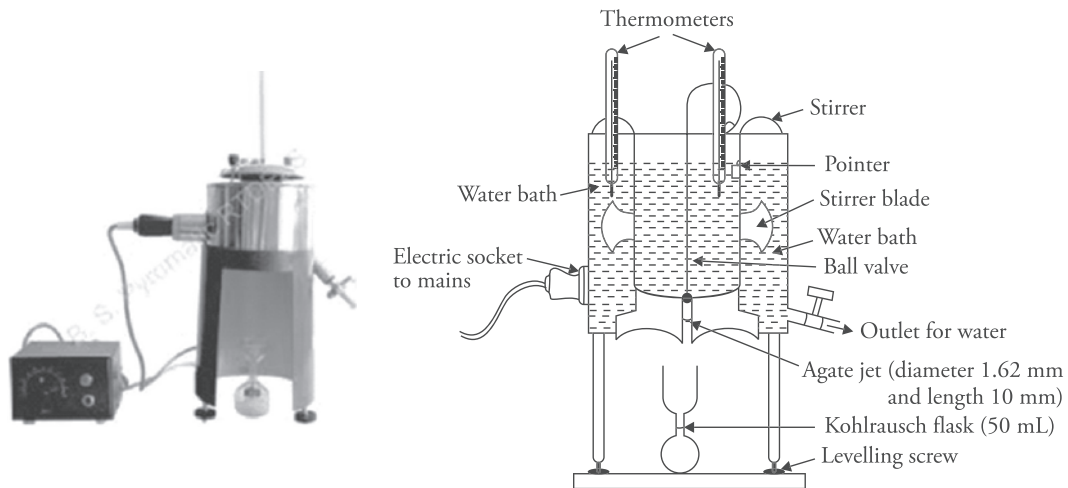


Figure 7.6 Redwood viscometer

- (i) **Oil cup** It is a silver plated brass cup (90 mm in height and 46.5 mm in diameter) with its upper end open. At the lower end, it has an orifice (length = 10 mm; diameter = 1.62 mm). The orifice is opened and closed with the help of a ball valve. The upper open end of the cup is closed with a lid having holes, one for the thermometer to record the temperature of the oil and the other for the ball valve. The side wall of the cup is provided with a pointer which is stout, tapered and pointing towards the upper side of the cup. The oil is filled till the tip of the pointer.
- (ii) **Heating bath** The oil is surrounded by a copper bath containing water. It is provided with a tap for emptying the water. It also bears an element for heating purposes. A thermometer is inserted to indicate the temperature of the water. The outer cylinder is also provided with a stirrer.
- (iii) **Spirit level** The apparatus rests on three legs provided with levelling screws at the bottom. The spirit level on the lid is used for levelling of the apparatus.
- (iv) **Kohlrausch flask** It is a calibrated flask placed at the bottom of the orifice used for collecting 50 mL of the oil sample.

Working The apparatus is levelled and oil is filled till the tip of the pointer taking care to close the orifice with the ball valve. The oil is heated. When the oil reaches the desired temperature, the ball valve is removed and the time taken for the flow of 50 mL of the oil sample is noted. This gives the redwood viscosity in seconds at the test temperature.

Conversion formula

The Redwood viscosity can be converted into kinematic viscosity using the formula

$$V = At - \frac{B}{t}$$

where V = kinematic viscosity of the oil in centistokes.

t = time of the flow in seconds

A and B are instrument constants. The value of

$A = 0.264, B = 190$ when $t = 40$ to 85 s

$A = 0.247, B = 65$ when $t = 85$ to 2000 s

Flash and fire point

Flash point is the minimum temperature at which the oil vapour–air mixture ignites for a moment, producing a spark when a small flame is brought near it. Fire point is the minimum temperature at which the vapours of the oil burn continuously for at least 5 s when a small flame is brought near it. Generally, the fire point is $5\text{--}40^\circ$ higher than the flash point. Flash and fire point is determined with the help of a *Pensky–Marten apparatus* (Fig. 7.7) which consists of the following parts.

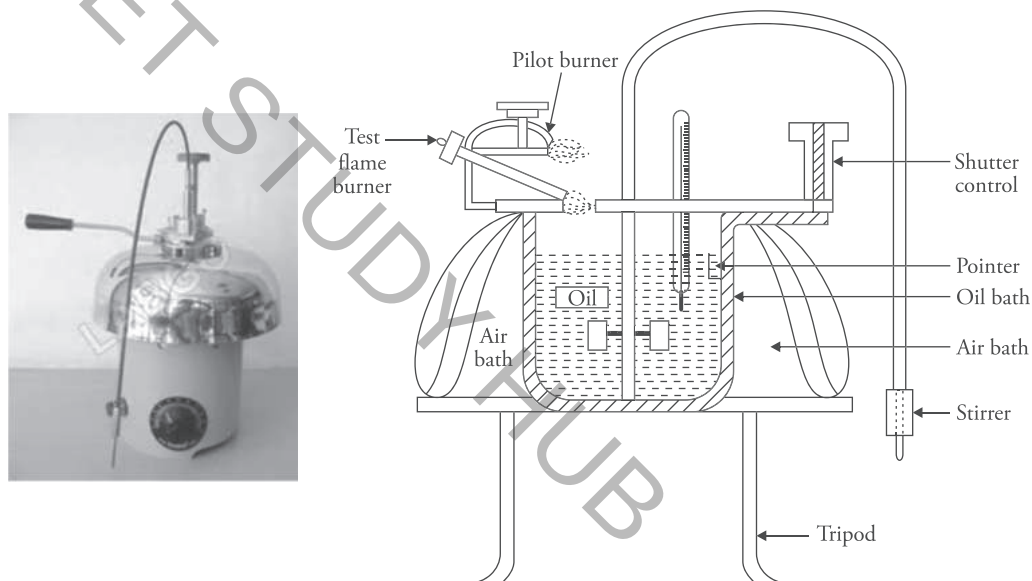


Figure 7.7 *Pensky–Marten apparatus*

- (i) **Oil cup** It is made of brass (5.5 cm deep and 5 cm in diameter). It has a lid provided with four openings of standard dimensions. One opening is for the thermometer, the second is for introducing the test flame, the third opening is for the stirrer and the fourth for the passage of air. The oil cup has a pointer on the side wall.
- (ii) **Shutter** It is provided at the top of the cup. By moving the shutter, the opening in the lid opens and the test flame is inserted inside the oil cup, thereby bringing it over the oil surface.
- (iii) **Air bath** The oil cup is surrounded by an air bath which is heated either by a gas burner or by an electric element.
- (iv) **Pilot burner** As the test flame is introduced in the opening, it extinguishes, but when it returns to its original position, it is automatically kindled by the pilot burner near the test flame.

Working The oil sample is filled in the cup till the tip of the pointer. A thermometer is inserted, the oil is heated and the stirrer is worked. The test flame is inserted inside the oil cup at every 1 °C rise in temperature. The temperature at which a flash (combination of light and sound) is observed is noted as the flash point and the temperature at which the flame continues to burn for at least 5 s is the fire point of the oil.

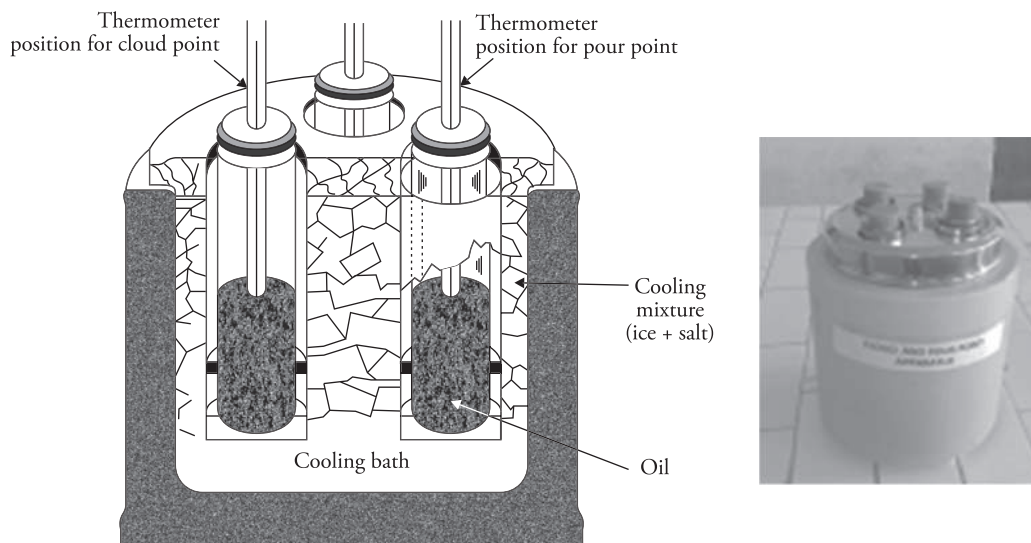
Significance Flash and fire point has no bearing on the lubricating properties of the oil but measures the fire hazards of the lubricants. A good lubricating oil should have flash and fire point at least 5–10° above the working temperature of the machine. It safeguards against the risk of fire during the operation of machine at high temperatures.

Cloud and pour point

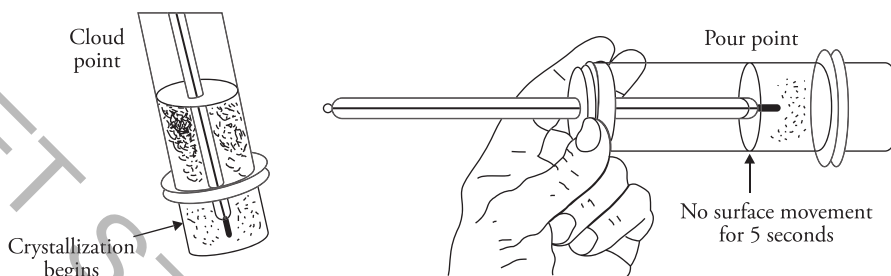
When oil is cooled slowly, the temperature at which it becomes cloudy or hazy is called the *cloud point* and the temperature at which it stops flowing when the tube is made horizontal (tilted) is called the *pour point*. Hence, the pour point refers to the maximum temperature at which a lubricant continues to flow. Below the pour point, the oil thickens and does not flow freely. Cloud and pour points can be determined with the help of a cloud and pour point apparatus.

Construction and working

The apparatus consists of a flat bottom tube (about 2 cm high and 3 cm in diameter) enclosed in an air jacket. The tube is half filled with oil and a thermometer is introduced in the oil. It is then placed in a freezing mixture (ice + NaCl or CaCl₂). As the oil cools, the temperature falls continuously. With every degree fall in temperature, the glass tube is withdrawn from the air jacket and observed. The temperature at which cloudiness or haziness appears is the cloud point of the lubricating oil (Fig. 7.8). After recording the cloud point, the tube is now withdrawn at every three degree fall of temperature. It is tilted and observed. The temperature at which the oil does not flow even when kept horizontal for at least 5 s is reported as the pour point of the lubricant (Fig. 7.8(b)).



(a) Cloud point measurement



(b) Pour point measurement

Figure 7.8 Cloud and pour point apparatus

Significance

Like flash and fire point, this too does not have a bearing on the lubricating properties of the oil. However, it signifies the suitability of the lubricant in cold conditions. Machines working in cold conditions should use a lubricant with low cloud and pour point, otherwise, the lubricant will solidify causing the jamming of the machine.

Aniline point

Aniline point of an oil is the temperature at which equal volumes of oil and aniline are just miscible. Aniline point is related to the percentage of aromatics in the oil. The lower the aniline point, the higher is the percentage of aromatics in the oil. Aromatic hydrocarbons are good solvents for natural rubbers and certain synthetic rubbers. The presence of these in oils will deteriorate rubber sealing and packing by dissolving them. A higher aniline point is therefore desirable.

To determine the aniline point, equal volumes of oil sample and aniline are mixed mechanically in a test tube and heated till a homogeneous solution is obtained. The tube is placed in a thermostat and cooled slowly. The temperature at which the two phases separate out is called the aniline point.

Steam emulsion number

When water enters an oil system, it forms an emulsion. These emulsions have poor lubricating properties and cause abrasion and wearing out of the lubricated parts of the machinery.

Hence, a good lubricating oil should form an emulsion with water, which breaks off quickly. The steam emulsion number (SEN) is determined by ASTM test. In this, 20 mL of oil is taken in a test tube and steam at 100 °C is bubbled in, till the temperature rises to 90° C. The tube is kept in a bath maintained at 90 °C and the time in seconds, when the oil and water separates out is noted. This time in seconds taken for oil and water to separate out into distinct layers is called an SEN. A good lubricant should have a low SEN.

Neutralisation number

It is an indication of the acidic and basic impurities in the lubricating oil. It is expressed either as a *total acid number* (TAN) which is the number of milligrams of KOH needed to neutralise any acid in one gram of oil or the *total base number* (TBN), which is the number of milligrams of HCl needed to neutralise any base in one gram of oil. Determination of acidity is more common and is expressed in terms of acid value or acid number. A known weight of oil is dissolved in a neutral alcohol and titrated against 0.1 N KOH using phenolphthalein as the indicator.

$$\text{Acid value} = \frac{\text{mL of KOH used} \times \text{normality} \times 56}{\text{Weight of the oil taken in g}}$$

Acid value of a lubricant should be less than 0.1. A value greater than 0.1 indicates that the oil has been oxidized.

Saponification number

An ester on alkaline hydrolysis gives alcohol and sodium or potassium salt of an acid. The saponification number is defined as the number of milligrams of KOH required to saponify the fatty acid present in 1 gram of oil. It is determined by refluxing a known quantity of oil with a known excess of potassium hydroxide solution and titrating the unused alkali against an acid.

Iodine value

It is a measure of the degree of unsaturation and the tendency of a fatty oil to absorb oxygen. Iodine value is defined as the number of milligrams of iodine absorbed by one gram of the oil. It is determined by adding a measured excess of a solution of iodine bromide in glacial acetic acid to a weighed quantity of oil dissolved in CCl₄ or acetic acid. The addition of the halogenating agent to the double bond is allowed to proceed to completion. The unused IBr is then back-titrated against a standard hypo solution using starch as an indicator.

Carbon residue

Lubricating oils contain high percentage of carbon in combined form. On being subjected to high temperature, they decompose and form a carbonaceous deposit. Oils which deposit a minimum amount of carbon are preferable. To determine the carbon residue, a bulb containing known weight of oil sample is heated to a temperature of 500 °C. The weight of the carbon residue formed is determined after cooling. The carbon residue is given by

$$\% \text{ carbon residue} = \frac{\text{Weight of residue in crucible/bulb}}{\text{Weight of original oil sample}} \times 100$$

7.8 Selection of Lubricants for Different Types of Machinery

In selecting a lubricant for a particular machine, various parameters have to be kept in mind. Lubricant selection is based on knowledge and experience. Tribology – the science of friction, lubrication and wear plays a vital role in the selection of lubricant. While selecting the lubricant, a lubrication engineer (tribo-engineer) should take into account the type of motion (sliding or rolling), speed of the machine, temperature condition, load (light or heavy) and the operating environment which includes moisture, air, oxygen, chemicals, liquids, vapours, etc.

Selection of lubricants for a few typical jobs is illustrated in Table 7.3.

Table 7.3 Selection of lubricants for a few typical jobs

Lubricant application	Properties of special interest in the lubricant	Recommended lubricant
Transformers	High resistivity, high dielectric strength, low-viscosity, high oxidation resistivity	Highly refined and moisture free mineral oils
Gears	Subjected to extreme pressure hence should have good oiliness, high film strength, oxidation resistance, high load carrying capacity	Thick mineral oils containing extreme pressure additives like metallic soaps, chlorine, sulphur or phosphorus compounds
Steam turbine	High-viscosity, high metal wetting, high temperature resistance, resistance to hydrolysis and oxidation	Steam refined paraffin base mineral oil compounded with fixed oil (vegetable and animal oils)
Internal combustion engines	High-viscosity index, high thermal stability	Petroleum oils containing additives
Delicate instruments like watches, clocks, sewing machines, scientific instruments	They are not exposed to high temperature, high load or water	Thin vegetable and animal oils like palm oil, neat oil, etc.

Cutting tools	Cool the tool, reduce wear, reduce power consumption by lubricating action, low-viscosity, prevent rust and fine finish	(i) <i>Heavy cutting</i> : Mineral oils containing additives like fatty oils, sulphurized fatty oils and chlorinated compounds (ii) <i>Light cuttings</i> : oil emulsions
Spindles in textile industry	Move at very high speeds	Thin oils containing additives like oxidation and rust inhibitors
Refrigeration system	Low pour point, low-viscosity, low cloud point	Naphthalene-based oils

Summary

- Lubricant is a substance introduced between two moving surfaces to reduce friction – a force which opposes the motion of any object.
- Apart from lubricating the moving surfaces, a lubricant acts as a coolant, reduces wear and tear, acts as a sealant for gases, reduces noise of the machine, reduces maintenance cost and increases the life of a machine.
- A good lubricant should have good oiliness (stick to the metal surface), its viscosity should be moderate, its ignition temperature should be high, freezing point should be low, it should not oxidize or hydrolyse in air.
- Lubrication which occurs by the formation of a thick film between the metal surfaces is called hydrodynamic or fluid film lubrication.
- In boundary lubrication, when the load is high and speed is slow, a thin film is formed between the metal surfaces. The film remains in its place due to adsorption on the metal.
- Extreme pressure lubrication occurs when extreme pressure additives like organic groups having chlorine, sulphur or phosphorus react with the metal surface forming a compound of high melting point.
- On the basis of their physical state, lubricants are classified as liquid lubricants, semi-solid lubricants and solid lubricants.
- Viscosity is the property by virtue which a liquid opposes its own flow. A good lubricant should have moderate viscosity and a viscosity which should not vary with temperature (viscosity index should be high).
- Viscosity is measured with the help of a Redwood viscometer and Saybolt viscometer.
- The temperature at which a lubricant produces flash is its flash point and the temperature at which it burns continuously for 5 s is called its fire point. Flash and fire points are measured by a Pensky–Martens apparatus. It signifies the fire hazards of a lubricant.

- The temperature at which a lubricant become cloudy is called its cloud point and the temperature at which it stops flowing is the pour point. They can be found with the help of a cloud and pour point apparatus. These points signify the suitability of a lubricant under cold conditions.
- Other important properties of lubricating oils are aniline point, emulsification, neutralisation number, saponification number, iodine value and carbon residue.

Review Questions

1. Define a lubricant. What are the functions of a lubricant and what are the qualities of a good lubricant?
2. What is hydrodynamic lubrication? Explain the phenomenon with the help of a suitable diagram.
3. How does lubrication occur by thin film or boundary lubrication? Distinguish between fluid film and boundary lubrication.
4. Write a short note on extreme pressure lubrication.
5. Classify lubricants.
6. What do you mean by blended or compounded oil? What are the various additives used to impart desirable properties to lubricants?
7. What are greases? Explain the different type of greases and also write down their advantage over liquid lubricants. What are the limitations of greases?
8. What are synthetic lubricants? Synthetic lubricants have an added advantage over natural lubricants. Explain.
9. What is viscosity? How will you determine the viscosity of a lubricating oil with the help of Redwood viscometer no. 1?
10. How does viscosity vary with temperature? Explain the determination of viscosity index of a lubricating oil. Suggest measures to improve the viscosity index of an oil sample.
11. What is flash and fire point? How can you determine the flash and fire point of a lubricant with the help of a Pensky–Marten apparatus?
12. Write short notes on
 - (i) Cloud and pour point
 - (ii) Emulsification
 - (iii) Aniline point
 - (iv) Neutralisation number
 - (v) Iodine value
 - (vi) Carbon residue
 - (vii) Saponification number.

Multiple Choice Questions

1. The functions of a lubricant are
 - (a) It reduces friction
 - (b) It acts as a coolant
 - (c) It acts as a sealant for gases
 - (d) All of the above
2. Delicate instruments like watches, clocks and light machines like scientific instruments and sewing machines have which type of lubrication
 - (a) Thick film or hydrodynamic lubrication
 - (b) Thin film or boundary lubrication
 - (c) High-pressure lubrication
 - (d) All the above
3. The type of lubrication under conditions of slow speed and high load is
 - (a) Thick film or hydrodynamic lubrication
 - (b) Thin film or boundary lubrication
 - (c) High-pressure lubrication
 - (d) All the above
4. Limitation of animal and vegetable oils is
 - (a) They have poor oiliness
 - (b) They oxidise to form gummy products
 - (c) They have extremely high-viscosity
 - (d) All the above
5. The common antioxidants used to retard the oxidation of lubricating oil is
 - (a) Aromatic phenolic or aminocompounds
 - (b) Alkylated naphthalene's
 - (c) Glycol
 - (d) Tricresylphosphate
6. Beside graphite the other commonly used solid lubricants are
 - (a) MoS_2
 - (b) Soapstone
 - (c) Talc
 - (d) All the above
7. As temperature increases the viscosity of a lubricating oil
 - (a) Increases
 - (b) Decreases
 - (c) Neither increases nor decreases
 - (d) Either increases or decreases
8. Viscosity index of a lubricating oil can be increased by adding organic polymers like
 - (a) Polyisobutylene
 - (b) Phenylenes
 - (c) Amines
 - (d) Ethers
9. The property of a lubricant that helps to safeguard against the risk of fire during operation of a machine
 - (a) Viscosity
 - (b) Flash and fire point
 - (c) Cloud and pour point
 - (d) Aniline point
10. The suitability of a lubricant in cold conditions is decided by
 - (a) Flash and fire point
 - (b) Steam emulsion number
 - (c) Cloud and pour point
 - (d) Iodine value

11. The parameter that helps in estimating the percentage of aromatics in oil is
(a) Aniline point (b) Iodine value
(c) Neutralisation number (d) Saponification number
12. The science of friction, lubrication and wear is called
(a) Tribology (b) Trichology
(c) Surface science (d) Lubrology

Solutions

- | | | | | | | |
|-------|-------|--------|--------|--------|-------|-------|
| 1 (d) | 2 (a) | 3 (b) | 4 (b) | 5 (a) | 6 (d) | 7 (b) |
| 8 (a) | 9 (b) | 10 (c) | 11 (a) | 12 (a) | | |

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